

Communication

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The density of surface coating can contribute to different antibacterial activities of gold nanoparticles

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KEYWORDS: Gold nanoparticles, Antibacterial, Surface ligand, Phenylboronic acid

## ABSTRACT

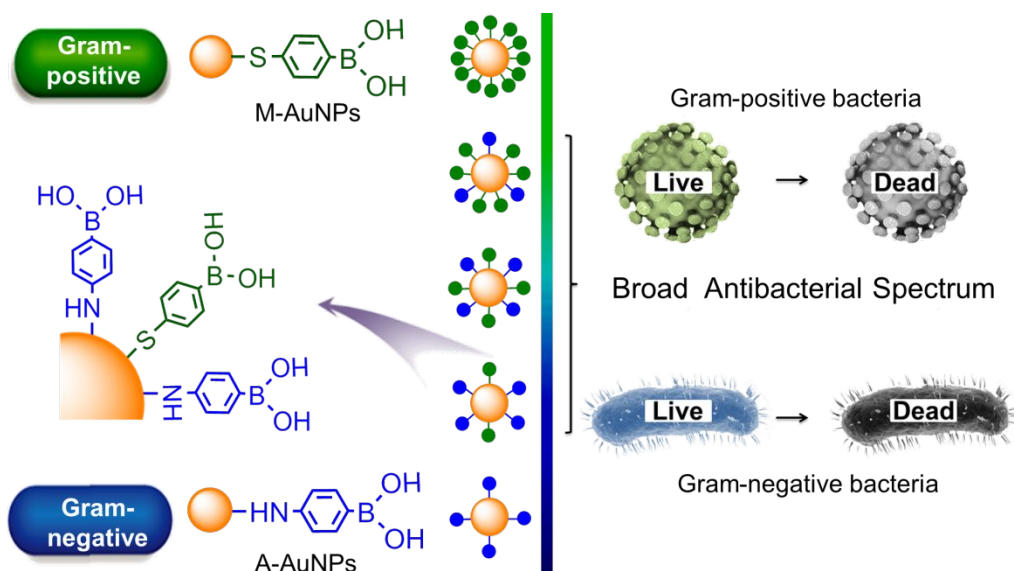
With the widespread use of antibiotics, the number of complex infection cases caused by unknown pathogens is increasing and novel antibiotics with tunable antibacterial spectra and low toxicity are highly desirable. Herein, we report that by selecting thiol or amine, two groups with different binding affinities with gold, as anchoring groups, phenylboronic acid can be decorated on gold nanoparticles (AuNPs) with different densities, which contributes to Gram-selective antibacterial activities of the AuNPs. The AuNPs modified with amine- or thiol-tethered phenylboronic acids specifically bind to lipopolysaccharide (LPS, Gram-negative) or lipoteichoic acid (LTA, Gram-positive) respectively. By modifying AuNPs with different ratios of thiol- and amine-tethered phenylboronic acids, the resulting AuNPs show potent and tunable antibacterial activity. The AuNPs-based antibacterial agents with optional Gram selectivity are promising for applications in personalized therapy.

## INTRODUCTION

The rising risk of antibiotic-resistant bacteria-caused complex infections has become a major global concern.<sup>1-2</sup> Thus, there is an urgent need for developing new generations of agents with optional antibacterial spectra to treat complex infections and avoid the generation of drug-resistance.<sup>3-4</sup> Nanomaterials are promising for achieving these goals. Functional nanomaterials<sup>5-6</sup> have been used to design various types of antimicrobials with broad<sup>7</sup> or tunable<sup>8</sup> antibacterial spectra. However, most of these nanomaterials have drawbacks. For instance, silver nanoparticles have high toxicity to host cells whereas the generation of reactive oxygen species by the Palladium nanocrystals may be harmful to the human body, thus greatly restricting their applications for the clinical use.

Gold nanoparticles (AuNPs) have become useful as potential antibiotics in recent years.<sup>9-10</sup> Compared with above nanomaterials, AuNPs allow for straightforward surface modification, which facilitates the development of new antibacterial combinations. Moreover, AuNPs have excellent biocompatibility which is the basic requirement for clinical applications. AuNPs themselves do not have antibacterial activity, however, when they are modified by certain small molecules which also have no antibacterial activity, the resulting AuNPs may show potent antibacterial activity, probably due to the combination of the multivalence and permeability of AuNPs and the interaction between modified AuNPs and bacterial surface molecules.<sup>11-12</sup> A recent study demonstrated that AuNPs covered with ligand shells comprising both positively and negatively charged ligands exhibited Gram-selective antibacterial action.<sup>13</sup> However, only narrow bacterial spectra were achieved.<sup>13</sup> To realize tunable Gram-selectivity of antibacterial effects of AuNPs in a facile way, we select candidate small molecules that have the potential to interact with the bacterial cell wall. One of the candidate molecules for such interaction is phenyl-boronic acid which is capable of binding with glycolipids through the boronic acid group,<sup>14-17</sup> because glycolipids are the major components of the bacteria cell wall. In addition, boric acid, the key functional group of phenylboronic acid, has disinfectant and antiseptic properties and could be used as a bacteriostatic agent. Thus, phenylboronic acid and its derivatives have great potential to be utilized to develop novel antibiotics, which has not been reported so far. Moreover, phenylboronic acid and its derivatives, which may bind with carbohydrates in the body and lead to cytotoxicity, were reported to have high biocompatibility.<sup>18</sup> Considering the function of phenylboronic acid and its derivatives, we attempt to decorate these molecules on AuNPs and explore if the resulting AuNPs have novel antibacterial activity.

We tether phenylboronic acid on AuNPs by two types of anchoring groups: -SH and -NH<sub>2</sub>, which can yield 2 kinds of phenylboronic acid derivatives and their isomers (mercaptophenylboronic acid, MBA, aminophenylboronic acid, ABA). Thiol and amine groups have different binding strengths to Au.<sup>19-22</sup> -SH is known to form strong S-Au covalent bonds whereas -NH<sub>2</sub> can form weak covalent bonds to Au.<sup>23-24</sup> We envisage that the steric effect of the benzene ring may compromise the chemical reactivity of the -SH and -NH<sub>2</sub> groups.<sup>25</sup> The different surface densities of ABA and MBA on AuNPs caused by the different binding affinities of Au-N and Au-S along with the difference between ABA-decorated AuNPs (A-AuNPs) and MBA-decorated AuNPs (M-AuNPs) as a whole, may elicit different antibacterial activities of the decorated AuNPs. As expected, A-AuNPs show potent antibacterial effects on Gram negative (G-) bacteria whereas M-AuNPs show potent antibacterial effects on Gram positive (G+) bacteria. By co-modification of MBA and ABA on the AuNPs with varying ratios, we obtain A/M-AuNPs with tunable antibacterial spectrum (Gram selectivity) (Figure 1).



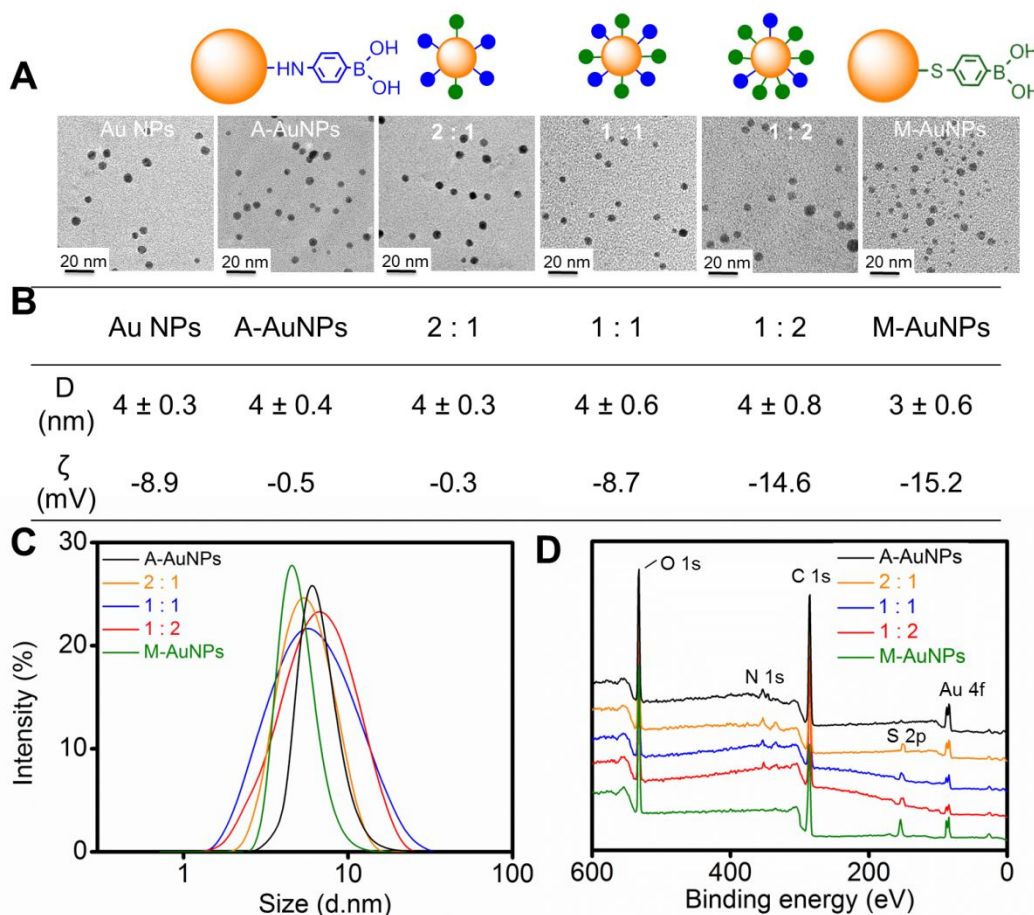
**Figure 1.** Schematic illustration of the preparation process of phenylboronic acid-derivatives-modified AuNPs by orchestrating the variation of ligands as tunable spectrum antibacterial agents.

RESULTS AND DISCUSSION

The particle size of the AuNPs capped with different ligands is  $4 \pm 0.4$  nm (A-AuNPs) and  $3 \pm 0.6$  nm (M-AuNPs) (Figure 2A-B). We select different G- and G+ bacteria as model bacteria (Table S1). Naked AuNPs and the individual components for synthesizing the AuNPs, including ABA and MBA, show no antibacterial activity (Figure S1). Even at an extremely high concentration ( $512 \mu\text{g/mL}$ ), ABA or MBA by themselves still shows no antibacterial effect (Table S2). Considering the above results, we do not use ABA or MBA by themselves as controls in the *in vivo* experiments to avoid the excessive use of animals. By measuring the minimum inhibitory concentration (MIC) of these nanoparticles, we find that A-AuNPs and M-AuNPs exhibit excellent antibacterial activity against G- and G+ bacteria respectively (Table 1, Table S3). Among the isomers of ABA and MBA, para-position ABA or MBA decorated Au NPs have the best antibacterial effect (Table S3), which are selected for subsequent experiments.

A-AuNPs and M-AuNPs are narrow spectrum antibacterial nanoparticles, which are incapable of treating complex infections caused by both G- and G+ bacteria. The different binding affinity of amine and thiol groups to Au means that the density of the phenylboronic acid on A-AuNPs and M-AuNPs are different. We thus hypothesize that we can obtain broad spectrum-AuNPs by co-modification of the AuNPs by ABA and MBA (A/M-AuNPs) and tuning the density of the ligands which could be achieved by simply adjusting the ratio of the ligands: ABA and MBA. For A/M-AuNPs with different molar ratios of ABA and MBA (2:1, 1:1, 1:2), the sizes are  $4 \pm 0.3$  nm,  $4 \pm 0.6$  nm and  $4 \pm 0.8$  nm respectively (Figure 2A-B). Zeta potential measurements

indicate that all the AuNPs are negatively charged (Figure 2B). Dynamic light scattering (DLS) curves indicate that there is no large aggregation in the particle solution (Figure 2C). The A/M-AuNPs are stable for long term storage at 4 °C (Figure S2). X-ray photoelectron spectroscopy (XPS) analysis indicates the simultaneous presence of Au (4f), C (1s), N (1s), and S (2p) electrons on A/M-AuNPs, indicating the successful functionalization of the molecules (ABA or MBA) on the AuNPs (Figure 2D). By calculating, we know the molar ratios of N to S of three kinds of A/M-AuNPs is 1.98:1, 0.98:1, and 0.52:1 respectively, which is near 2:1, 1:1, and 1:2 respectively (Table S4).



**Figure 2.** Characterization of phenylboronic acid-derivative-modified AuNPs. (A) TEM images of the AuNPs with different ligand modifications. (B) Size distribution and Zeta potential of the AuNPs modified by different ligands. (C) DLS analysis of the Au NPs modified by different

ligands. (D) Chemical components of the phenylboronic acid-derivative-modified AuNPs analysed by XPS.

Interestingly, A/M-AuNPs, modified by two different ligands, show gradually tunable antibacterial spectrum and effects according to the molar ratio of ABA/MBA on the surface of the nanoparticles. For the G- bacteria, with the increase of ABA ratio, A/M-AuNPs show increased antibacterial effect against G- bacteria (Table 1, Figure S3A-B). For G+ bacteria, with the increase of MBA ratio, A/M-AuNPs show increased antibacterial effect against G+ bacteria (Table 1, Figure S3C-D). As the A/M-AuNPs (2:1) group has optimized antibacterial activity in inhibiting both G+ and G- bacteria, we select it for the following experiments.

| Materials  |       | Gram-negative bacteria |                    |             |                 |             |                 |
|------------|-------|------------------------|--------------------|-------------|-----------------|-------------|-----------------|
|            |       | <i>E. coli</i>         | MDR <i>E. coli</i> | <i>P. a</i> | MDR <i>P. a</i> | <i>K. p</i> | MDR <i>K. p</i> |
| A-AuNPs    |       | 2                      | 6                  | 6           | 7               | 12          | 4               |
|            | 2 : 1 | 3                      | 6                  | 3           | 6               | 6           | 2               |
| A/M-AuNPs  | 1 : 1 | 3                      | 6                  | 6           | 12              | 12          | 6               |
|            | 1 : 2 | 6                      | 9                  | 12          | 49              | 12          | 12              |
| M-AuNPs    |       | >64                    | >64                | >64         | >64             | >64         | >64             |
| Ampicillin |       | 4                      | >64                | >64         | 32              | >64         | >64             |
|            |       | Gram-positive bacteria |                    |             |                 |             |                 |
|            |       | <i>S. a</i>            | MDR <i>S. a</i>    | <i>S. e</i> | MDR <i>S. e</i> | <i>E. f</i> | MDR <i>E. f</i> |
| A-AuNPs    |       | >64                    | >64                | >64         | >64             | >64         | >64             |
|            | 2 : 1 | 6                      | 6                  | 4           | 13              | 4           | 12              |
| A/M-AuNPs  | 1 : 1 | 6                      | 6                  | 4           | 13              | 4           | 12              |
|            | 1 : 2 | 6                      | 6                  | 3           | 12              | 3           | 12              |
| M-AuNPs    |       | 6                      | 6                  | 2           | 13              | 3           | 12              |



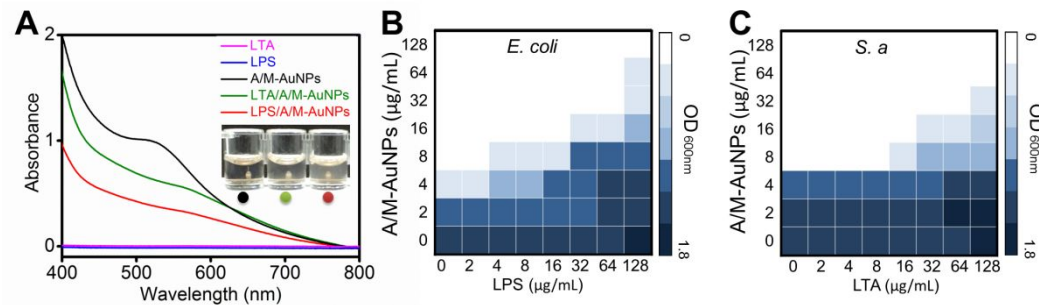
|            |   |     |     |     |   |     |
|------------|---|-----|-----|-----|---|-----|
| Ampicillin | 1 | >64 | >64 | >64 | 4 | >64 |
|------------|---|-----|-----|-----|---|-----|

**Table 1** Antibacterial activity of the AuNPs modified by different ligands indicated by MIC (μg/mL). Details of the bacterial strains are listed in Table S1.

We characterize the morphological change of AuNPs-treated bacteria by scanning electron microscope (SEM) and transmission electron microscope (TEM). The phenylboronic acid derivatives-modified AuNPs-treated bacteria merge together (white arrows), implying the disruption of the bacterial structures (Figure S4A). The phenylboronic acid derivatives-modified AuNPs can adhere on the bacteria cell wall (marked in red arrows) and locate at the cytosol of the bacteria (Figure S4B), this means that the AuNPs can enter the bacteria. SYTO 9/PI <sup>26</sup> staining shows that the phenylboronic acid derivatives-modified AuNPs can increase the permeability of the bacteria cell wall (Figure S4C-D, Figure S5). Furthermore, the membrane potential measurement indicates the change in the membrane potential which may lead to a decline in cellular metabolism <sup>27</sup> (Figure S6). From above results, we can conclude that the phenylboronic acid derivatives-modified AuNPs can indeed induce the destruction of the bacteria cell wall.

To explore if A/M-AuNPs kill bacteria by interacting with specific surface molecules of the bacteria, we carry out blockage assays <sup>28</sup> by introducing typical surface components of G- (*e.g.*, LPS <sup>29</sup> from *E. coli* O111:B4 or *K. p.*, Figure S7A) and G+ (*e.g.*, LTA <sup>30</sup> from *S. a* or *Bacillus subtilis*, Figure S7B) bacteria into the nanoparticle solutions before interacting with the bacteria. The addition of purified LPS or LTA induces the change of the A/M-AuNPs solution from complete transparency to high turbidity rapidly (Figure S7C-D), which means that the nanoparticles may form aggregations. We select LPS from *E. coli* O111:B4 or LTA from *S. a* in the following experiments. The characteristic UV-Vis absorption of A/M-AuNPs is around 520

nm, which disappears with the addition of LPS or LTA (Figure 3A). Obvious aggregations of LPS/A/M-AuNPs or LTA/A/M-AuNPs are formed when A/M-AuNPs are incubated with LPS or LTA respectively (Figure S8), indicating the potent interactions between A/M-AuNPs and the bacterial polysaccharides. Using the same method, we find that A-AuNPs and M-AuNPs can form aggregates with LPS and LTA respectively (Figure S9). We observe the influence of dosage of LPS or LTA on the antibacterial activity of LPS or LTA-modified A/M-AuNPs. Increased concentrations of LPS dramatically inhibits the antibacterial activity of A/M-AuNPs against *E. coli* (Figure 3B). Similarly, LTA decreases the antibacterial activity of A/M-AuNPs against *S. a* in a concentration dependent manner (Figure 3C). The above results reflect that LPS and LTA are the potential binding targets of A/M-AuNPs to G- or G+ bacteria respectively.



**Figure 3.** A/M-AuNPs exert bactericidal activity by interacting with LPS and LTA. (A) UV-vis spectra of the LPS, LTA, A/M-AuNPs, LPS/A/M-AuNPs, and LTA/A/M-AuNPs. The antibacterial activity of A/M-AuNPs at different concentrations against *E. coli* or *S. a* in the presence of LPS and LTA at concentrations ranging from 0 to 128 µg/mL is plotted as diagram (B) and (C) respectively.

We further confirm the interaction of the AuNPs and polysaccharides by carrying out isothermal titration calorimetry (ITC) measurements. LPS and A-AuNPs show stronger calorimetric response compared with M-AuNPs or A/M-AuNPs group (Figure S10A-C, Table S5). The binding affinity between LTA and M-AuNPs (Figure S10D-F) is stronger than those of

the other two groups (A-AuNPs or A/M-AuNPs). Thus, the results of ITC provide the evidence of the direct interactions between the AuNPs and the key structural molecules of the bacteria. The ITC results also demonstrate that the loosely coated AuNPs (A-AuNPs) tend to interact with LPS, whereas the densely coated Au NPs (M-AuNPs) are prone to interact with LTA, which is consistent with the antibacterial spectra of them.

We further explore if the variation of LPS or LTA could influence the nano-particle-bacteria interactions. Both rough LPS (rLPS) and smooth LPS (sLPS) induce obvious aggregation of A/M-AuNPs, indicating the potent interaction between the A/M-AuNPs and the LPS from different bacterial species (Figure S11). Disk diffusion assay (Figure S11C-D) demonstrates that O-side chain of LPS may play roles during the A/M-AuNPs-bacterial interactions. We also compare the interactions of A/M-AuNPs with LTA from different bacteria (*S. a* and *Bacillus subtilis*, *B. s*). The addition of both types of LTA induces aggregation of the A/M-AuNPs solution (Figure S12).

Collectively, the interaction of A-AuNPs with LPS, M-AuNPs with LTA, or A/M-AuNPs with both LPS and LTA on the bacteria cell wall may trigger the bacteriolysis and the downstream destructive effects on bacterial metabolism which needs further investigation in future studies.

To further explore the possible mechanism underlying the differential behavior of A-AuNPs and M-AuNPs. We calculate the number of ABA and MBA molecules modified per unit area of the Au NPs, which is 29.99 for A-AuNPs and 55.66 for M-AuNPs. We also carry out XPS experiments which show approximately 2 times of Au:S ratio compared with the Au:N ratio (Table S6). This means that ABA and MBA have differential capabilities to tether to the surface of the AuNPs. That is to say, the MBA molecules on the AuNPs may densely distribute and pack together, whereas ABA molecules on the AuNPs may have a sparse distribution. In comparison,

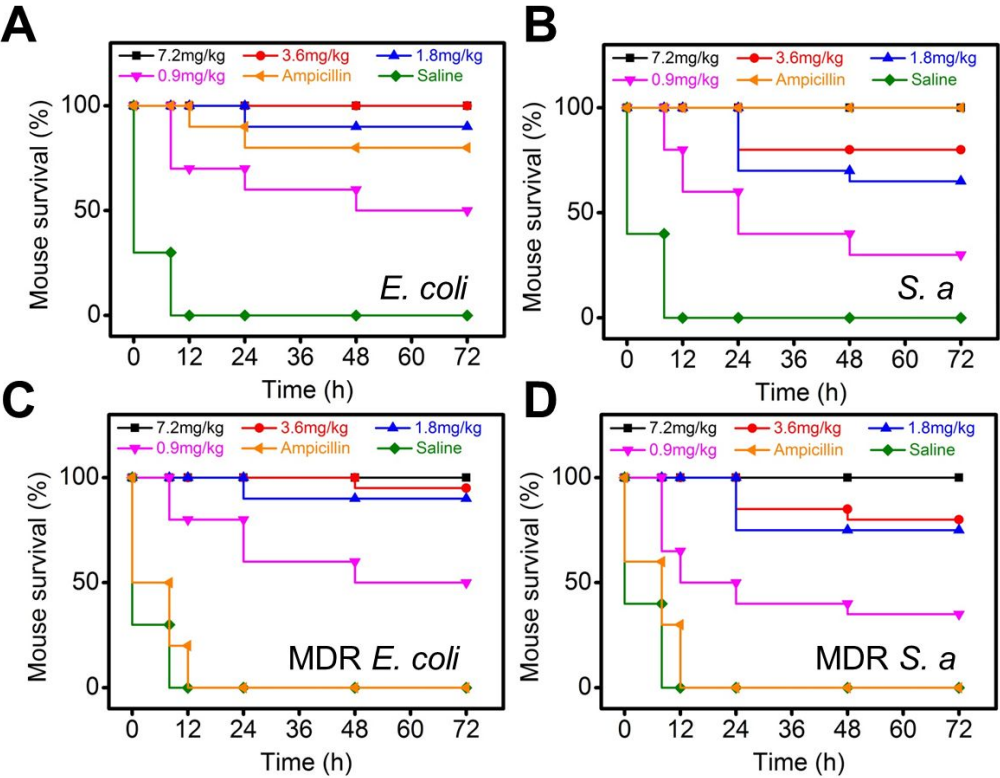
ABA&MBA molecules on A/M-AuNPs may have an intermediate density. We also characterize the typical adsorption peak of the AuNPs by UV-Vis spectrum (Figure S13). The absorption peak of bare AuNPs is at 520 nm, the absorption peak of A-AuNPs at 520 nm is as strong as that of bare AuNPs, whereas the absorption of the M-AuNPs is drastically weakened (Figure S13). We further analyze the element distribution of individual particles by energy-dispersive spectroscopy (EDS) mapping. The density of MBA on the gold nanoparticles is 2.84 times that of ABA (Figure S14), indicating that the ratio of elemental sulfur to gold is significantly higher than the ratio of nitrogen to gold. XPS data also indicate that the ligand density on M-AuNPs is larger than that on A-Au NPs (Table S6). Thus, the density of ABA and MBA on the Au NPs is different, which dramatically influence the behaviors of the decorated nanoparticles.

To explore if changing the concentration of precursor phenylboronic acid derivatives with one kind of anchoring groups can induce different grafting densities of the Au NPs, we synthesize A-AuNPs and M-AuNPs with different concentrations of ABA or MBA as substrates respectively using the one-pot synthesis protocol. However, as the concentration of the molecule decreases, the synthesized M-AuNPs gradually become much bigger (8 nm, volume increasing 8 times), which prevents a straightforward comparison between the M-AuNPs with different ligand densities. If we increase the concentration of MBA from 5 mM to 15 mM, the Zeta potential of the synthesized M-AuNPs shows insignificant variation, but their size becomes smaller (Figure S15, Figure 2), which will inevitably influence the bioactivity of the nanoparticles. To achieve high ligand density of A-AuNPs, we increase the concentration of ABA from 5 mM to 15 mM (3 times of the concentration for synthesizing A-AuNPs). The increase of the concentration of ABA does not bring the apparent change of the Zeta potential (Figure S15, Figure 2), indicating that increasing the ABA concentration cannot achieve denser surface ligand. By contrast, if we

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3 decrease the ABA concentration, the synthesized A-AuNPs show increased aggregation (Figure.  
4 S15). Thus, it is infeasible to achieve different ligand densities by using just one kind of  
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6 anchoring groups but different amounts of precursor during the synthesis. Considering that the  
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8 alteration of the particle states can dramatically influence the antibacterial activity of the AuNPs,  
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10 we are unable to study the ligand density-induced variation of antibacterial spectra under a  
11  
12 comparable condition.  
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17 We also explore if we can obtain AuNPs with different surface modifications (ABA or MBA)  
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19 by using ligand exchange reaction on AuNPs synthesized with sodium citrate/HAuCl<sub>4</sub> reduction  
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21 method (Figure S16). The addition of 5 mM ABA or MBA (the same amount for the one step-  
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23 method) into the AuNPs synthesized with citrate reduction method induces immediate color  
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25 change of the solutions (Figure S16A-B), implying the aggregation of the AuNPs. To avoid  
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27 aggregation, we dilute the ABA or MBA (0.2  $\mu$ M or 2  $\mu$ M) solution for the ligand exchange.  
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29 Still, the AuNPs aggregate to different extents as shown by both macroscopic observation and  
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31 TEM images (Figure S16C-D, Figure S17). Considering the drastically altered physiochemical  
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33 property of the AuNPs synthesized by the ligand exchange method, we cannot explore the  
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35 antibacterial mechanism using this kind of AuNPs.  
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42 The comparison between the different AuNPs synthesis methods indicates that the most  
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44 effective way to keep the identical AuNPs size but different surface ligand densities is to obtain  
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46 AuNPs using different ratios of ABA and MBA with different binding affinities to Au as  
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48 substrates by one-pot synthesis method.  
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**Figure 4.** *In vivo* evaluation of the therapeutic effects of A/M-AuNPs on intra-abdominal infected mice. Survival ratios of the (A) *E. coli*, (B) *S. a*, (C) MDR *E. coli*, and (D) MDR *S. a* infected mice treated by different concentrations of A/M-AuNPs, Ampicillin, and saline are calculated and analyzed.

We evaluate the therapeutic effect of A/M-AuNPs *in vivo* using acute intra-abdominal infection model on mice. Without antibacterial treatment (saline), all the mice die at 12 h after *E. coli* infection. By contrast, with the treatment of A/M-AuNPs (360  $\mu\text{g/mL}$ , 3.6 mg/kg), all the infected mice survive 72 h after *E. coli* infection, while only 80 % of the mice treated with same concentration of ampicillin survive after 72 h (Figure 4A). For MDR *E. coli* infection, A/M-AuNPs (720  $\mu\text{g/mL}$ ) lead to 100 % mice survival after 72 h, on the contrary, all the ampicillin-treated mice die after 12 h (Figure 4C). For the *S. a* infection, A/M-AuNPs (720  $\mu\text{g/mL}$ ) and the same concentration of ampicillin can maintain 100 % bacterial infected mice survival (Figure 4B). For the MDR *S. a* infection, all the ampicillin-treated mice die after 12 h, by contrast, A/M-

AuNPs (720  $\mu\text{g/mL}$ ) lead to 100 % mice survival after 72 h (Figure 4D). These results demonstrate the prominent antibacterial effect of A/M-AuNPs for drug-resistant bacteria infections *in vivo*. By comparison, AuNPs alone does not have any antibacterial activity (Figure S18). We evaluate the median lethal dose ( $\text{LD}_{50\text{i.v.}}$ ) of A/M-AuNPs in mice. At an extremely high concentration (920 mg/kg), only 50 % of the mice treated by A/M-AuNPs die within 12 h (Figure S19). In this *in vivo* assay, the dose of A/M-AuNPs used in the treatment (7.2 mg/kg) is less than 1 % of the  $\text{LD}_{50\text{i.v.}}$  (920 mg/kg), indicating the excellent antimicrobial activity and excellent biosafety of A/M-AuNPs. Furthermore, compared with the commercially available antibacterial agents, A/M-AuNPs have the best biosafety (Table S7).

## CONCLUSIONS

In this study, by modulating the density of surface ligands (phenylboronic acid) on AuNPs by tuning the ratio of different anchoring groups, amine or thiol group, we synthesize the Au NPs with a tunable-spectrum antibacterial capability. A-AuNPs and M-AuNPs could specifically target LPS on G- bacteria and LTA on G+ bacteria respectively, which can explain the tunable antibacterial spectrum of A/M-AuNPs. Moreover, the extremely high  $\text{LD}_{50\text{i.v.}}$  (920 mg/kg) of A/M-AuNPs confirms the high biosafety of the nanoparticles. Nevertheless, the different types of anchoring groups may have other important roles of the Gram selectivity, which needs to be explored in the future. Collectively, A/M-AuNPs show the great potential as a candidate for clinical applications, especially for treating complex, MDR bacteria-induced infections.

## ASSOCIATED CONTENT

**Supporting Information**

Methods and supplementary Figures (S1-S19) and Tables (S1-S7).

The Supporting Information is available free of charge at <http://pubs.acs.org>.

Synthesis and characterization of nanoparticles, experimental procedures for the cell studies and animal studies, and more supplementary figures and tables.

Figure S1-S19 and Table S1-S7.

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**Author Contributions**

Le Wang, Junchuan Yang, Wenfu Zheng, and Xingyu Jiang designed the experiment, Le Wang, Sixiang Li, Jiaxiang Yin, Junchuan Yang, and Qizhen Li performed the experiments. Le Wang, Wenfu Zheng, and Xingyu Jiang analyzed the data and wrote the manuscript. Shaoqin Liu, Wenfu Zheng and Xingyu Jiang supervised the whole project.

**Notes**

The authors declare no competing financial interest.



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## ABBREVIATIONS

AuNPs, gold nanoparticles; MBA, mercaptophenylboronic acid; ABA aminophenylboronic acid; G<sup>+</sup>, Gram positive; G<sup>-</sup>, Gram negative; LTA, lipoteichoic acid; LPS, lipopolysaccharide; sLPS, smooth lipopolysaccharide; rLPS, rough lipopolysaccharide; MIC, minimum inhibitory concentration; XPS, X-ray photoelectron spectroscopy; LD<sub>50i.v.</sub>, median lethal dose.

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